LAB RECORD

23CSE111-Object Oriented Programming

Submitted by

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND

ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Lab Record work for 23CSE111- Object Oriented Programming Subject submitted by CH.SC.U4CSE24027 – LOMADA SAHITHI in “Computer Science and Engineering” is a Bonafide record of the work carried out under my guidance and supervision at Amrita School of Computing, Chennai.

This lab examination held on: 08/04/2025

Course Incharge

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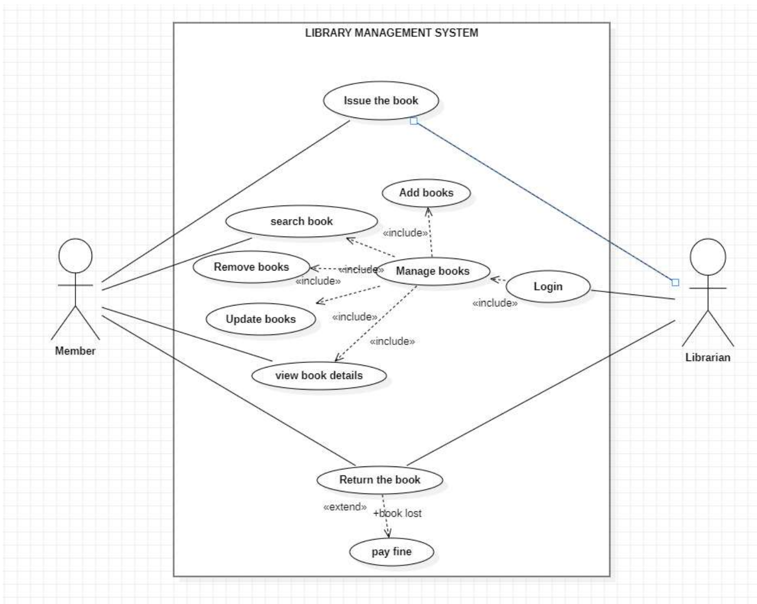
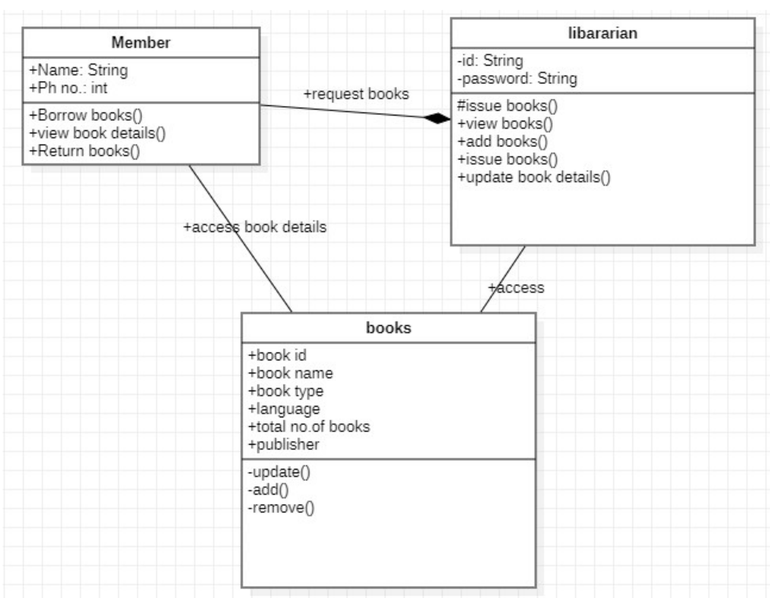
UML DIAGRAMS

1: LIBRARY MANAGEMENT SYSTEM

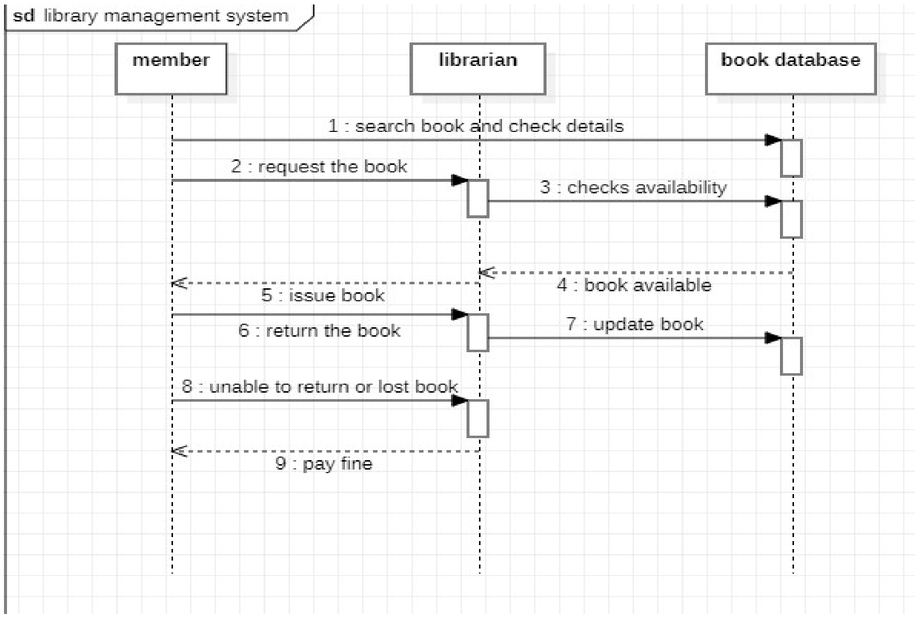
1A-USECASE DIAGRAM:



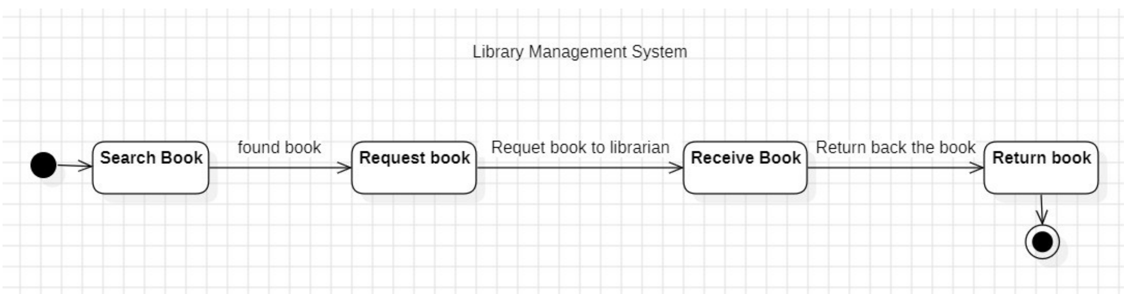
1B:CLASS DIAGRAM:



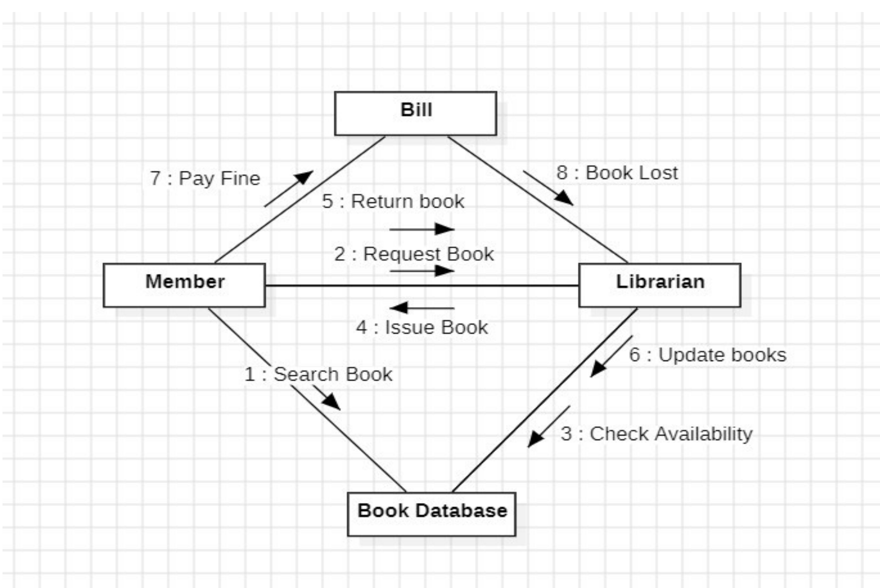
1C-SEQUENCE DIAGRAM:



1D: STATE DIAGRAM

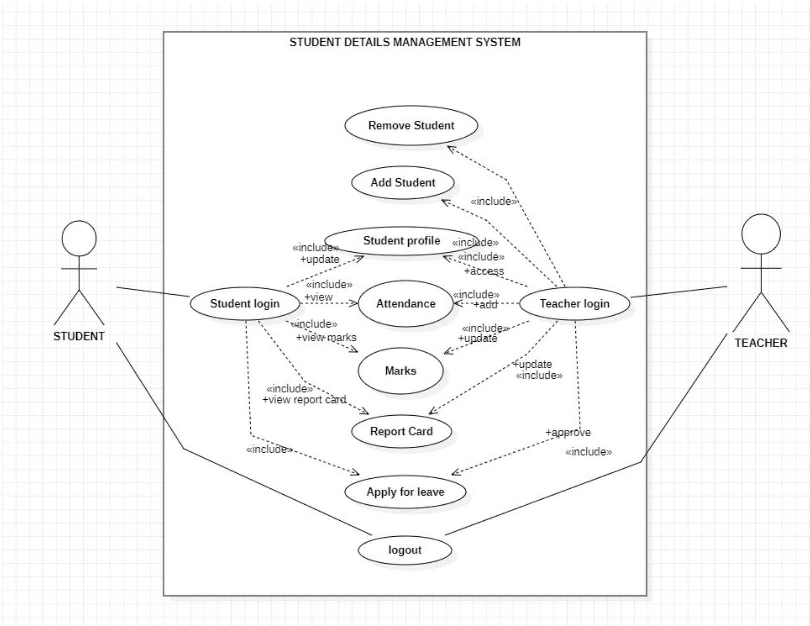


1E: COLLABORATION DIAGRAM

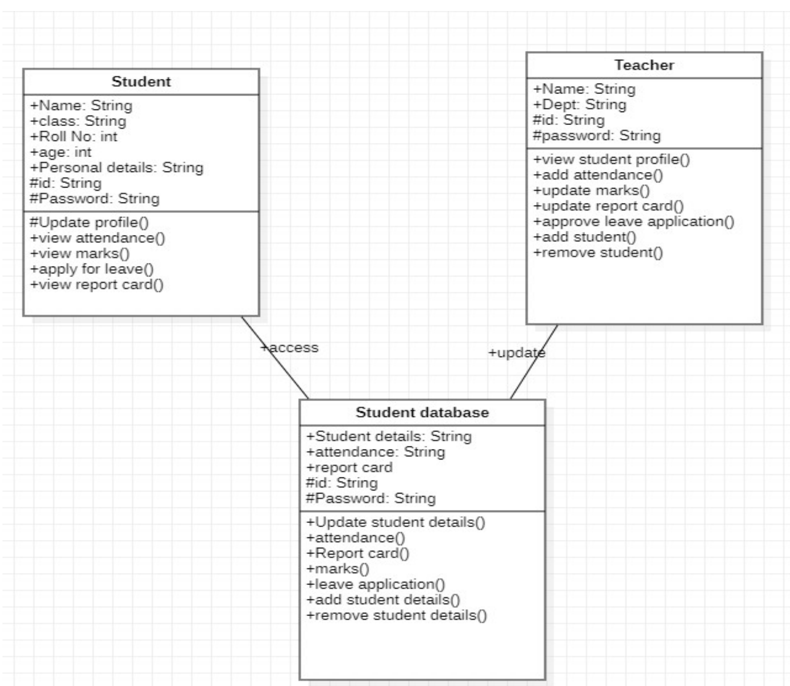


2- STUDENT DATABASE MANAGEMENT SYSTEM

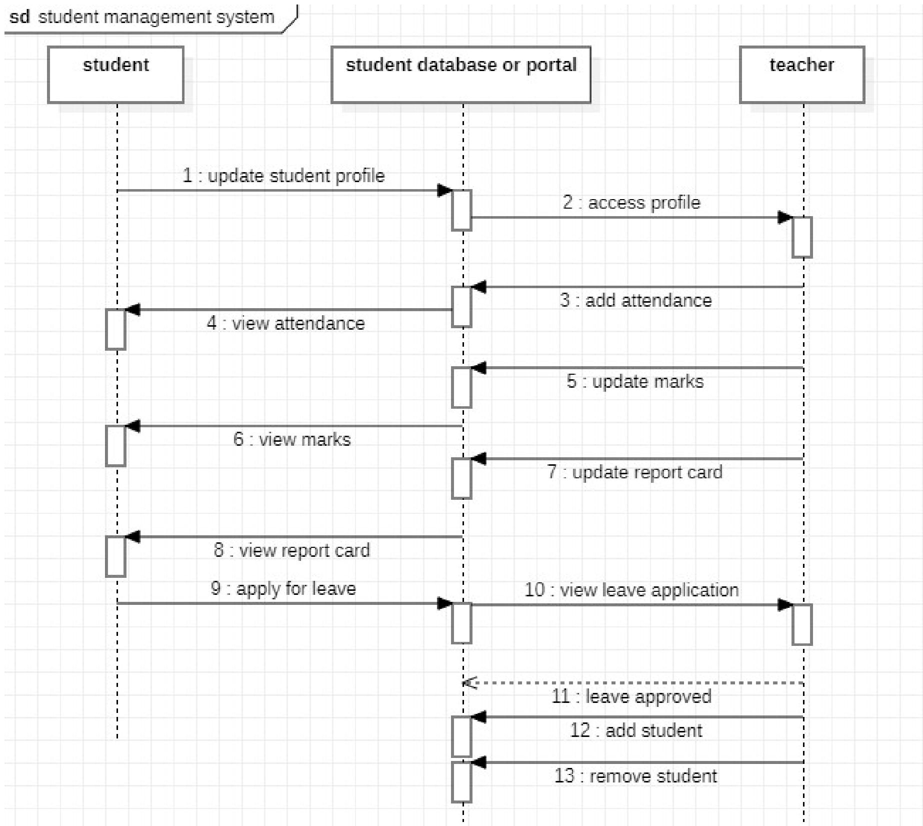
2A-USECASE DIAGRAM



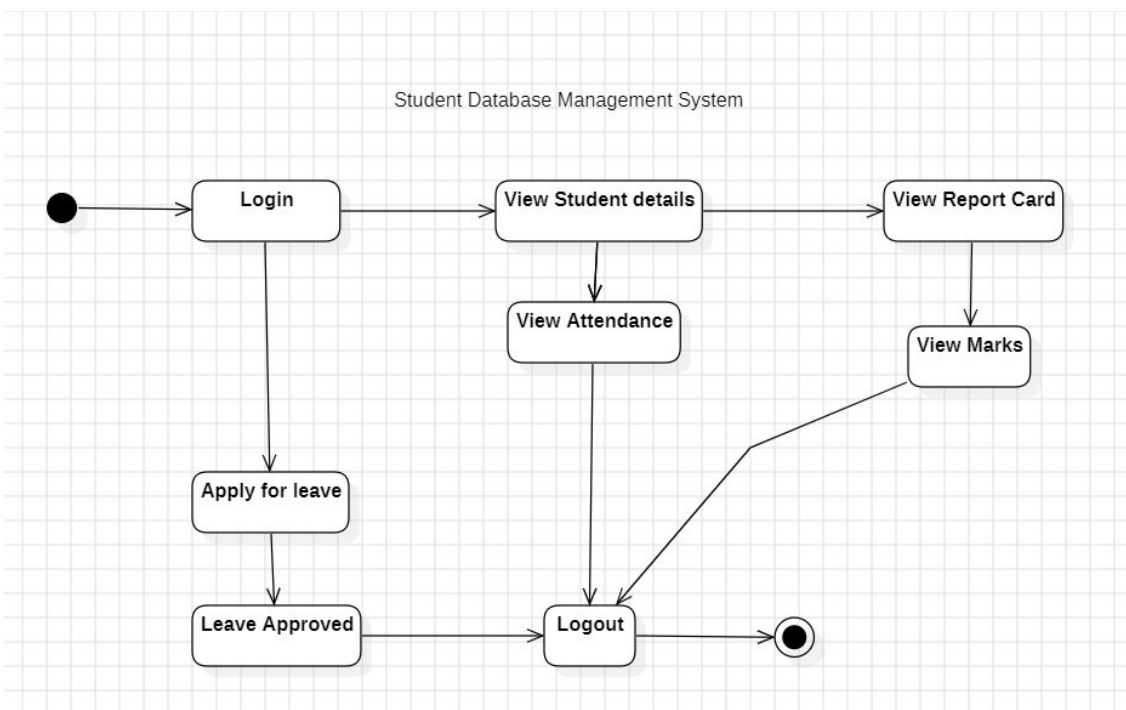
2B-CLASS DIAGRAM



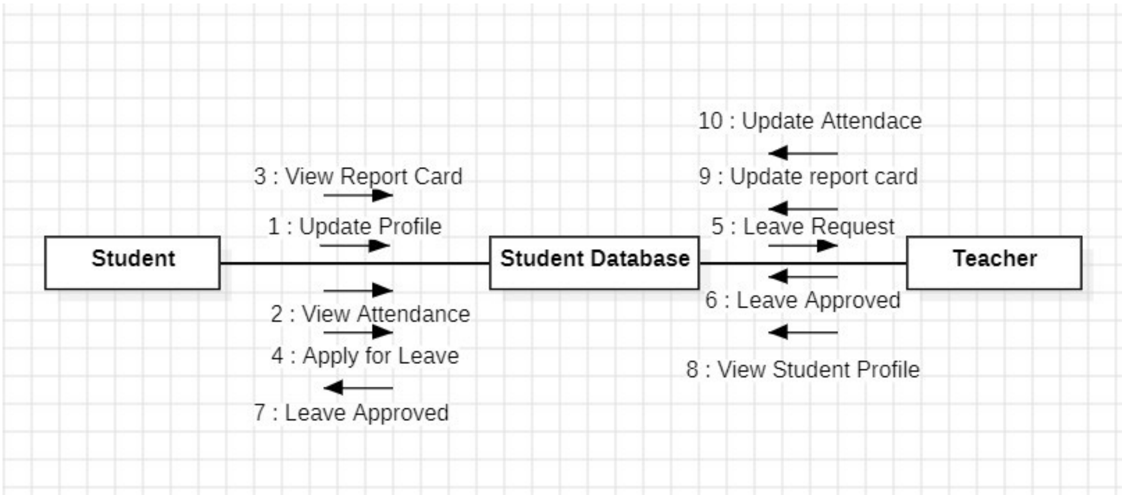
2C-SEQUENCE DIAGRAM



2D: STATE DIAGRAM



2E: COLLABORATION DIAGRAM



BASIC JAVA PROGRAMS

3A- Sum of digits of a number

CODE:

```
//sum of digits of a number
import java.util.Scanner;
public class p1{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter a number");
    int a=sc.nextInt();
    int sum=0;
    while(a>0){
      int n=a%10;
      sum=sum+n;
      a=a/10;
    }
  System.out.println(sum);
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p1.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p1
Enter a number
123
6
```

3B- Greatest of three numbers

CODE:

```
// greatest of three Numbers
import java.util.Scanner;
public class p2{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter a number");
    int a= sc.nextInt();
    System.out.println("Enter a number");
    int b= sc.nextInt();
    System.out.println("Enter a number");
    int c=sc.nextInt();
    if(a>b && a>c){
       System.out.println("Greatest number is:"+ a);
    }
    else if(b>a && b>c){
      System.out.println("Greatest number is:"+ b);
    }
    else{
      System.out.println("Greatest number is:"+ c);
    }
  }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p2.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p2
Enter a number
12
Enter a number
10
Enter a number
11
Greatest number is:12
```

3C- Factorial

CODE:

```
//factorial
import java.util.Scanner;
public class p3{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter a number");
    int a= sc.nextInt();
    int m=1;
    for(int i=1; i<a+1; i++){
       m=m*i;
    }
  System.out.println("Factorial is:"+ m);
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p3.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p3
Enter a number
5
Factorial is:120
```

3D- Check palindrome or not

CODE:

```
//check if given string is palindrome or not
import java.util.Scanner;
public class p4{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    String b= "";
    System.out.println("Enter a string");
    String s= sc.nextLine();
    int n=s.length();
    String s1= s.toLowerCase();
    for(int i=n-1; i>=0; i--){
      b = b + s1.charAt(i);
    }
    if(s1.equals(b)){
      System.out.println("It's a pallindrome");
    }
    else{
      System.out.println("It's not a pallindrome");
    }
  }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p4.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p4
Enter a string
madam
It's a pallindrome
```

3E- Armstrong Number

CODE:

```
//Armstong Number
import java.util.Scanner;
public class p5{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter a number");
    int n= sc.nextInt();
    int n1=n;
    int a=0;
    while(n>0){
      int t=n%10;
      a=(int) (Math.pow(t,3))+a;
      n=n/10;
    }
  if(a==n1){
    System.out.println("It's an Armstrong Number");
  }
  else{
    System.out.println("It's not an Armstrong Number");
  }
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p5.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p5
Enter a number
512
It's not an Armstrong Number
```

3F- Reverse a number

CODE:

```
// reverse a number
import java.util.Scanner;
public class p6{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter a number");
    int a=sc.nextInt();
    int rev=0;
    while(a>0){
      int n=a%10;
      rev=rev*10+n;
      a=a/10;
    }
  System.out.println(rev);
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p6.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p6
Enter a number
1234
4321
```

3G: Program using break statement

CODE:

```
//break statement
import java.util.Scanner;
public class p7{
  public static void main(String[] args){
    Scanner s= new Scanner(System.in);
    System.out.println("Enter the number to break the loop in the range 1-10:");
    int o=s.nextInt();
    System.out.println("The numbers are:");
    for (int i = 1; i <= 10; ++i) {
       if (i == o){
         break;
       }
       System.out.println(i);

     }
   }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p7.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p7
Enter the number to break the loop in the range 1-10:
4
The numbers are:
1
2
3
```

3H- Skip numbers using continue statement

CODE:

```
import java.util.Scanner;
public class p8{
  public static void main(String[] args){
    Scanner s= new Scanner(System.in);
    System.out.println("Enter the number to skip in the range of 1-10:");
    int o=s.nextInt();
    System.out.println("The numbers are:");
    for(int i=0; i<=10; i++){
      if(i==o){
        continue;
      }
      System.out.println(i);
    }
  }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p8.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p8
Enter the number to skip in the range of 1-10:
5
The numbers are:
0
1
2
3
4
6
7
8
9
10
```

3I- Determine vowel or not using switch case

CODE:

```
import java.util.Scanner;
public class p9{
  public static void main(String[] args){
    Scanner input = new Scanner(System.in);
      char ch;
      System.out.printf("Enter a Character : ");
      ch = input.next().charAt(0);
      if((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')){
          switch (ch){
          case 'A':
            case 'E':
            case 'I':
            case 'O':
            case 'U':
            case 'a':
            case 'e':
            case 'i':
            case 'o':
            case 'u':
                  System.out.println("This is a Vowel");
                      break;
            default:
                      System.out.println("This is a Consonant");
                      break;
        }
     }
     else{
        System.out.println("This is Not an alphabet");
     }
   }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p9.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p9
Enter a Character : e
This is a Vowel
```

3J- Student grading using do-while loop

CODE:

```
import java.util.Scanner;
public class p10{
  public static void main(String[] args){
    boolean result;
    Scanner s= new Scanner(System.in);
    do{
       System.out.println("Enter the student avg");
       float avg=s.nextFloat();

       if(avg >=90)
         System.out.println("Grade A");
       else if(avg>=80 && avg<90)
         System.out.println("Grade B");
       else if(avg>=60 && avg<80)
         System.out.println("Grade C");
       else
         System.out.println("Grade D");

    System.out.println("Student belong to Sec-A true/false");
    result= s.nextBoolean();
   }while(result);

}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac p10.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java p10
Enter the student avg
80
Grade B
Student belong to Sec-A true/false
false
```

INHERITANCE

SINGLE INHERITANCE:

4A: Calculate salary of employees
CODE:

```
import java.util.Scanner;
class Employee{
double salary;
Employee(double salary){
this.salary=salary;
}
}
class Programmer extends Employee{
double bonus;
Programmer(double bonus, double salary){
super(salary);
this.bonus=bonus;
}
public void totalSalary(){
System.out.println(salary+bonus);
}
}
public class Record{
public static void main(String[] args){
Scanner s=new Scanner(System.in);
System.out.println("Enter the Salary: ");
double salary=s.nextDouble();
System.out.println("Enter the bonus: ");
double bonus=s.nextDouble();
Programmer p=new Programmer(salary,bonus);
p.totalSalary();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the Salary:
12000
Enter the bonus:
5000
17000.0
```

4B: Display Bank Account details

CODE:

```
import java.util.Scanner;
class Account {
    String name;
    String AccNo;
    double bal;
    Account(String name, String AccNo, double bal) {
        this.name = name;
        this.AccNo = AccNo;
        this.bal = bal;
    }
    public void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Account Number: " + AccNo);
        System.out.println("Balance: " + bal);
    }
}
class BankAccount extends Account {
    BankAccount(String name, String AccNo, double bal) {
        super(name, AccNo, bal);
    }
    public void withdraw(double amount) {
        if (bal >= amount) {
            bal -= amount;
            System.out.println("Balance after withdrawal: " + bal);
        } else {
            System.out.println("Insufficient Balance");
```

```
}
    }
    public void deposit(double amount) {
        bal += amount;
        System.out.println("Balance after deposit: " + bal);
    }
}
public class Record {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        System.out.println("Enter Name: ");
        String name = s.nextLine();
        System.out.println("Enter Account Number: ");
        String AccNo = s.nextLine();
        System.out.println("Enter Balance in your account: ");
        double bal = s.nextDouble();
        BankAccount b = new BankAccount(name, AccNo, bal);
        b.displayDetails();
        System.out.println("Enter the amount to withdraw:");
        double withdrawAmount = s.nextDouble();
        b.withdraw(withdrawAmount);
        System.out.println("Enter the amount to deposit:");
        double depositAmount = s.nextDouble();
        b.deposit(depositAmount);
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter Name:
Lomada Sahithi
Enter Account Number:
123
Enter Balance in your account:
3000
Name: Lomada Sahithi
Account Number: 123
Balance: 3000.0
Enter the amount to withdraw:
100
Balance after withdrawal: 2900.0
Enter the amount to deposit:
120
Balance after deposit: 3020.0
```

MULTILEVEL INHERITANCE:

5A: Calculate Semester wise average

CODE:

```
import java.util.Scanner;
class sem1{
double avg1;
sem1(double avg1){
this.avg1=avg1;
}
}
class sem2 extends sem1{
double avg2;
sem2(double avg1, double avg2){
super(avg1);
this.avg2=avg2;
}
}
class sem3 extends sem2{
double avg3;
sem3(double avg1, double avg2, double avg3){
super(avg1, avg2);
this.avg3=avg3;
}
}
class sem4 extends sem3{
double avg4;
sem4(double avg1, double avg2, double avg3, double avg4){
super(avg1, avg2, avg3);
this.avg2=avg2;
this.avg3=avg3;
this.avg4=avg4;
}
public void OverallAvg(){
System.out.println("Overall avg of the student after sem4: "+(avg1+ avg2+
avg3+ avg4)/4);
}
}
public class Record{
public static void main(String[] args){
```

```
Scanner s=new Scanner(System.in);
System.out.println("Enter sem1 avg: ");
double avg1=s.nextDouble();
System.out.println("Enter sem2 avg: ");
double avg2=s.nextDouble();
System.out.println("Enter sem3 avg: ");
double avg3=s.nextDouble();
System.out.println("Enter sem4 avg: ");
double avg4=s.nextDouble();
sem4 a= new sem4(avg1, avg2, avg3, avg4);
a.OverallAvg();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter sem1 avg:
90
Enter sem2 avg:
90
Enter sem3 avg:
95
Enter sem4 avg:
85
Overall avg of the student after sem4: 90.0
```

5B: Calculate Area and Volume of House

CODE:

```
import java.util.Scanner;
class House{
double length;
double width;
House(double length, double width){
this.length=length;
this.width=width;
}
}
class Area extends House{
Area(double length, double  width){
super(length,width);
```

```
}
public double AreaofHouse(){
return(length*width);
}
}
class Volume extends Area{
double height;
Volume(double length, double width, double height){
super(length,width);
this.height=height;
}
public double volume(){
return(length*width*height);
}
}
public class Record{
public static void main(String[] args){
Scanner s=new Scanner(System.in);
System.out.println("Enter the length of the house:");
double length=s.nextDouble();
System.out.println("Enter the width of the house:");
double width=s.nextDouble();
System.out.println("Enter the height of the house:");
double height=s.nextDouble();
Volume v=new Volume(length, width, height);
System.out.println("Area of the House: "+v.AreaofHouse());
System.out.println("Volume of the HOuse: "+v.volume());
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the length of the house:
100
Enter the width of the house:
100
Enter the height of the house:
100
Area of the House: 10000.0
Volume of the HOuse: 1000000.0
```

HIERARICHICAL INHERITANCE

6A: Add bonus to the base salary of the employees

CODE:

```
import java.util.Scanner;
class Employee{
double baseSalary;
Employee(double baseSalary){
this.baseSalary=baseSalary;
}
public void display(){
System.out.println("Base Salary: "+baseSalary);
}
}
class Manager extends Employee{
double bonus;
Manager(double baseSalary, double bonus){
super(baseSalary);
this.bonus=bonus;
}
public void Salary(){
System.out.println("Total Salary: "+(baseSalary+bonus));
}
}
class HR extends Employee{
double bonus;
HR(double baseSalary, double bonus){
super(baseSalary);
this.bonus=bonus;
}
public void Salary(){
System.out.println("Total Salary: "+(baseSalary+bonus));
}
}
public class Record{
public static void main(String[] args){
Scanner s= new Scanner(System.in);
System.out.println("Enter the base salary for Manager: ");
double b1=s.nextDouble();
```

```
System.out.println("Enter the bonus for Manager: ");
double bonus1=s.nextDouble();
System.out.println("Enter the base salary for HR: ");
double b2=s.nextDouble();
System.out.println("Enter the bonus for HR: ");
double bonus2=s.nextDouble();
Manager m=new Manager(b1, bonus1);
HR h=new HR(b2, bonus2);
System.out.println("Manager: ");
m.display();
m.Salary();
System.out.println("HR: ");
h.display();
h.Salary();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the base salary for Manager:
12000
Enter the bonus for Manager:
3000
Enter the base salary for HR:
20000
Enter the bonus for HR:
5000
Manager:
Base Salary: 12000.0
Total Salary: 15000.0
HR:
Base Salary: 20000.0
Total Salary: 25000.0
```

6B: Display Employee details

CODE:

```
import java.util.Scanner;
class Employee {
String name;
int empId;
Employee(String name, int empId) {
this.name = name;
this.empId = empId;
}
public void display() {
System.out.println("Employee ID: " + empId );
System.out.println("Name: " + name);
}
}
class Developer extends Employee {
String language;
Developer(String name, int empId, String language) {
super(name, empId);
this.language = language;
}
public void work(){
System.out.println("Programming language known: "+language);
}
}
class Manager extends Employee {
int teamSize;
Manager(String name, int empId, int teamSize) {
super(name, empId);
this.teamSize = teamSize;
}
public void manager(){
System.out.println("Team size: "+teamSize);
}
}
public class Record{
public static void main(String[] args){
Scanner s=new Scanner(System.in);
System.out.println("Enter the name of the developer: ");
```

```
String n1=s.nextLine();
System.out.println("Enter employee id: ");
int i1=s.nextInt();
s.nextLine();
System.out.println("Entet the language known: ");
String l=s.nextLine();
System.out.println("Enter the name of the Manager: ");
String n2=s.nextLine();
System.out.println("Enter employee id: ");
int i2=s.nextInt();
System.out.println("Enter the team size: ");
int t=s.nextInt();
Developer d= new Developer(n1,i1,l);
System.out.println("Developer Details");
d.display();
d.work();
System.out.println();
Manager m = new Manager(n2,i2,t);
System.out.println("Manager details");
m.display();
m.manager();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the name of the developer:
Sahithi
Enter employee id:
123
Entet the language known:
Java
Enter the name of the Manager:
Sanjana
Enter employee id:
121
Enter the team size:
6
Developer Details
Employee ID: 123
Name: Sahithi
Programming language known: Java

Manager details
Employee ID: 121
Name: Sanjana
Team size: 6
```

HYBRID INHERITANCE

7A: Display details of Employees

CODE:

```
class Employee{
String name;
int empId;
double salary;
Employee(String name, int empId, double salary){
this.name = name;
this.empId = empId;
this.salary = salary;
}
void showDetails(){
System.out.println("Employee ID: " + empId);
System.out.println("Name: " + name);
System.out.println("Salary: " + salary);
}
}
class Developer extends Employee{
String language;
Developer(String name, int empId, double salary, String language){
super(name, empId, salary);
this.language = language;
}
void writeCode(){
System.out.println("Language known:" + language);
}
}
class Manager extends Employee {
int teamSize;
Manager(String name, int empId, double salary, int teamSize) {
super(name, empId, salary);
this.teamSize = teamSize;
}
void manageTeam(){
System.out.println(name + " is managing a team of " + teamSize + "
employees.");
}
}
```

```
class TechLead extends Developer{
TechLead(String name, int empId, double salary, String language){
super(name, empId, salary, language);
}
void leadProject() {
System.out.println(name + " is leading the project and coding in " + language);
}
}
public class Record {
public static void main(String[] args) {
Developer d = new Developer("Ram", 101, 75000, "Java");
d.showDetails();
d.writeCode();
System.out.println();
Manager m = new Manager("Sahithi", 201, 90000, 10);
m.showDetails();
m.manageTeam();
System.out.println();
TechLead l = new TechLead("Sanjana", 301, 100000, "Python");
l.showDetails();
l.writeCode();
l.leadProject();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Employee ID: 101
Name: Ram
Salary: 75000.0
Language known:Java

Employee ID: 201
Name: Sahithi
Salary: 90000.0
Sahithi is managing a team of 10 employees.

Employee ID: 301
Name: Sanjana
Salary: 100000.0
Language known:Python
Sanjana is leading the project and coding in Python
```

7B: Display details of students

CODE:

```
class Person{
String name;
int age;
Person(String name, int age){
this.name = name;
this.age = age;
}
void showDetails(){
System.out.println("Name: " + name);
System.out.println("Age: " + age);
}
}
class Student extends Person {
String course;
Student(String name, int age, String course){
super(name, age);
this.course = course;
}
void study(){
System.out.println("Course: "+ course);
}
}
class Teacher extends Person{
String subject;
Teacher(String name, int age, String subject) {
super(name, age);
this.subject = subject;
}
void teach(){
System.out.println("Subject: "+subject);
}
}
class GraduateStudent extends Student {
String researchTopic;
GraduateStudent(String name, int age, String course, String researchTopic) {
super(name, age, course);
```

```
this.researchTopic = researchTopic;
}
void conductResearch(){
System.out.println("Research Topic: "+researchTopic);
}
}
public class Record{
public static void main(String[] args) {
Student s= new Student("Sahithi", 20, "Computer Science");
System.out.println("Student:");
s.showDetails();
s.study();
System.out.println();
System.out.println("Teacher:");
Teacher t = new Teacher("Sanjana", 40, "Mathematics");
t.showDetails();
t.teach();
System.out.println();
GraduateStudent g= new GraduateStudent("Samyuktha", 25, "Physics",
"Quantum Mechanics");
System.out.println("Research:");
g.showDetails();
g.study();
g.conductResearch();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Student:
Name: Sahithi
Age: 20
Course: Computer Science

Teacher:
Name: Sanjana
Age: 40
Subject: Mathematics

Research:
Name: Samyuktha
Age: 25
Course: Physics
Research Topic: Quantum Mechanics
```

POLYMORPHISM

8A-CONTRUCTOR PROGRAM: Calculate area and volume of the room

CODE:

```
import java.util.Scanner;
class Room{
  double length;
  double width;
  Room(double length, double width){
    this.length=length;
    this.width=width;
    }
   void area(){
     System.out.println(length*width);
   }
}
class Height extends Room{
   double height;
   Height(double length, double width, double height){
     super(length, width);
     this.height=height;
   }
   void vol(){
     System.out.println(length*width*height);
   }
}

public class practice{
  public static void main(String args[]){
    Scanner s= new Scanner(System.in);
    System.out.println("Enter length");
    double length=s.nextDouble();
    System.out.println("enter width");
    double width=s.nextDouble();
    System.out.println("Enter Height");
    double height=s.nextDouble();
    Height p= new Height(length, width, height);
```

```
    p.area();
    p.vol();
  }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac practice.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java practice
Enter length
10
enter width
10
Enter Height
10
100.0
1000.0
```

9A-CONTRUCTOR OVERLOADING PROGRAM: Display details of different employees

CODE:

```
class Employee {
String name;
int age;
double salary;
Employee(String name) {
this.name = name;
this.age = 0;
 this.salary = 0.0;
 }
 Employee(String name, int age) {
 this.name = name;
 this.age = age;
 this.salary = 0.0;
 }
 Employee(String name, int age, double salary) {
 this.name = name;
 this.age = age;
 this.salary = salary;
 }
void display() {
```

```
System.out.println("Employee Name: " + name + ", Age: " + age + ", Salary: " +
salary);
    }
}
public class Record{
    public static void main(String[] args) {
        Employee emp1 = new Employee("Sahithi");
        Employee emp2 = new Employee("Sahithi", 20);
        Employee emp3 = new Employee("Sahithi", 20, 50000);
        emp1.display();
        emp2.display();
        emp3.display();
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Employee Name: Sahithi, Age: 0, Salary: 0.0
Employee Name: Sahithi, Age: 20, Salary: 0.0
Employee Name: Sahithi, Age: 20, Salary: 50000.0
```

METHOD OVERLOADING PROGRAMS:

10A: Calculate area of shapes

CODE:

```
import java.util.Scanner;
class Shape{
public int area(int a, int b){
 return(a*b);
 }
public double area(double a){
 return(3.14*a*a);
 }
 }
public class Record{
public static void main(String[] args){
Scanner s= new Scanner(System.in);
System.out.println("Enter length of Rectangle:");
```

```
int l=s.nextInt();
System.out.println("Enter width of the Rectangle:");
int w=s.nextInt();
System.out.println("Enter radius of circle: ");
double r=s.nextDouble();
Shape a=new Shape();
System.out.println("Area of the Rectangle: "+a.area(l,w));
System.out.println("Area of the circle: "+a.area(r));
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter length of Rectangle:
10
Enter width of the Rectangle:
20
Enter radius of circle:
10
Area of the Rectangle: 200
Area of the circle: 314.0
```

10B: Add bonus to the employee's salary

CODE:

```
import java.util.Scanner;
class Employee{
public double Salary(double Salary){
return(Salary);
}

public double Salary(double Salary, double Bonus){
return(Salary+Bonus);
}
}
public class Record{
public static void main(String[] args){
Scanner s=new Scanner(System.in);
System.out.println("Enter the Salary: ");
double sal=s.nextDouble();
System.out.println("Enter the Bonus: ");
double bonus=s.nextDouble();
```

```
Employee e=new Employee();
System.out.println("Salary: "+e.Salary(sal));
System.out.println("Salary + Bonus: "+e.Salary(sal, bonus));
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the Salary:
20000
Enter the Bonus:
5000
Salary: 20000.0
Salary + Bonus: 25000.0
```

METHOD OVERRIDING PROGRAM

11A: Calculate area of circle and rectangle

CODE:

```
class Shape {
double calculateArea() {
return 0;
}
}
class Circle extends Shape {
double radius;
Circle(double radius) {
this.radius = radius;
}
double calculateArea() {
return Math.PI * radius * radius;
}
}
class Rectangle extends Shape {
double width;
double height;
Rectangle(double width, double height) {
this.width = width;
this.height = height;
}
```

```
double calculateArea() {
return width * height;
}
}
public class Record {
public static void main(String[] args) {
Circle circle = new Circle(10);
System.out.println("Area of Circle: " + circle.calculateArea());
Rectangle rectangle = new Rectangle(10, 20);
System.out.println("Area of Rectangle: " + rectangle.calculateArea());
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Area of Circle: 314.1592653589793
Area of Rectangle: 200.0
```

11B: Display Employee details

CODE:

```
import java.util.Scanner;
class Employee{
String name;
double salary;
Employee(String name, double salary){
this.name=name;
this.salary=salary;
}
public void display(){
System.out.println("Name: "+name);
System.out.println("Salary: "+salary);
}
}
class Manager extends Employee{
Manager(String name, double salary){
super(name, salary);
}
public void display(){
```

```
System.out.println("Name: "+name);
System.out.println("Salary: "+salary);
}
}

public class Record{
public static void main(String[] args){
Scanner s=new Scanner(System.in);
System.out.println("Enter name of the employee: ");
String name=s.nextLine();
System.out.println("Enter salary of the employee: ");
double sal=s.nextDouble();
s.nextLine();
System.out.println("Enter name of the Manager: ");
String name1=s.nextLine();
System.out.println("Enter salary of the Manager: ");
double sal1=s.nextDouble();
Employee e=new Employee(name, sal);
Manager m=new Manager(name1, sal1);
System.out.println("Employee details:");
e.display();
System.out.println();
System.out.println("Manager details: ");
m.display();

}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter name of the employee:
Sahithi
Enter salary of the employee:
30000
Enter name of the Manager:
Sanjana
Enter salary of the Manager:
35000
Employee details:
Name: Sahithi
Salary: 30000.0

Manager details:
Name: Sanjana
Salary: 35000.0
```

ABSTRACTION

INTERFACE PROGRAMS:

12A: Calculate perimeter and area of shapes

CODE:

```
import java.util.Scanner;
 interface Shape {
   public double area();
   public double perimeter();
}
class Square implements Shape {
   double side;

   Square(double side) {
      this.side = side;
   }
   public double area() {
      return (side * side);
   }
   public double perimeter() {
      return (4 * side);
   }
}
class Rectangle implements Shape {
   double length;
   double breadth;

   Rectangle(double length, double breadth) {
      this.length = length;
      this.breadth = breadth;
   }
   public double area() {
      return (length * breadth);
   }
    public double perimeter() {
      return (2 * (length + breadth));
   }
}
```

```
public class Record {
  public static void main(String[] args) {
      Scanner s = new Scanner(System.in);
      System.out.println("Enter the side of the square: ");
      double side = s.nextDouble();
      System.out.println("Enter length of the rectangle: ");
      double length = s.nextDouble();
      System.out.println("Enter the breadth of the rectangle: ");
      double breadth = s.nextDouble();
      Square sq = new Square(side);
      System.out.println("\nSquare:");
      System.out.println("Area: " + sq.area());
      System.out.println("Perimeter: " + sq.perimeter());
      Rectangle rec = new Rectangle(length, breadth);
      System.out.println("\nRectangle:");
      System.out.println("Area: " + rec.area());
      System.out.println("Perimeter: " + rec.perimeter());
  }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the side of the square:
10
Enter length of the rectangle:
10
Enter the breadth of the rectangle:
20

Square:
Area: 100.0
Perimeter: 40.0

Rectangle:
Area: 200.0
Perimeter: 60.0
```

12B: Add bonus to the salary and display employee details

CODE:

```
import java.util.Scanner;
interface Employee {
   void salary(double bonus);
```

```
    void displayDetails();
}
class Manager implements Employee {
    double salary;
    String name;
    int empId;
    int teamSize;
    Manager(double salary, String name, int empId, int teamSize) {
        this.salary = salary;
        this.name = name;
        this.empId = empId;
        this.teamSize = teamSize;
    }
    public void salary(double bonus) {
        System.out.println("Base Salary: " + salary);
        System.out.println("Bonus: " + bonus);
        System.out.println("Total Salary: " + (salary + bonus));
    }
    public void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + empId);
        System.out.println("Team Size: " + teamSize);
    }
}
class Programmer implements Employee {
    double salary;
    String name;
    int empId;
    String language;

    Programmer(double salary, String name, int empId, String language) {
        this.salary = salary;
        this.name = name;
        this.empId = empId;
        this.language = language;
    }
     public void salary(double bonus) {
        System.out.println("Base Salary: " + salary);
        System.out.println("Bonus: " + bonus);
        System.out.println("Total Salary: " + (salary + bonus));
```

```
    }
    public void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + empId);
        System.out.println("Language Known: " + language);
    }
}
public class Record{
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        System.out.println("Name of the Manager: ");
        String n1 = s.nextLine();
        System.out.println("Enter Employee ID: ");
        int i1 = s.nextInt();
        System.out.println("Enter Team Size: ");
        int t = s.nextInt();
        System.out.println("Enter Salary of the Manager: ");
        double sal1 = s.nextDouble();
        System.out.println("Enter Bonus for Manager: ");
        double b1 = s.nextDouble();
        s.nextLine(); // Consume newline
        System.out.println("Name of the Programmer: ");
        String n2 = s.nextLine();
        System.out.println("Enter Employee ID: ");
        int i2 = s.nextInt();
        s.nextLine(); // Consume newline
        System.out.println("Enter Language Known: ");
        String l = s.nextLine();
        System.out.println("Enter Salary of the Programmer: ");
        double sal2 = s.nextDouble();
        System.out.println("Enter Bonus for Programmer: ");
        double b2 = s.nextDouble();
        Employee m = new Manager(sal1, n1, i1, t);
        Employee p = new Programmer(sal2, n2, i2, l);
        System.out.println("\nManager Details:");
        m.displayDetails();
        m.salary(b1);
        System.out.println("\nProgrammer Details:");
        p.displayDetails();
        p.salary(b2);
```

```
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Name of the Manager:
Sahithi
Enter Employee ID:
123
Enter Team Size:
5
Enter Salary of the Manager:
30000
Enter Bonus for Manager:
2000
Name of the Programmer:
Dimple
Enter Employee ID:
124
Enter Language Known:
Python
Enter Salary of the Programmer:
25000
Enter Bonus for Programmer:
5000

Manager Details:
Name: Sahithi
Employee ID: 123
Team Size: 5
Base Salary: 30000.0
Bonus: 2000.0
Total Salary: 32000.0

Programmer Details:
Name: Dimple
Employee ID: 124
Language Known: Python
Base Salary: 25000.0
Bonus: 5000.0
Total Salary: 30000.0
```

12C: Calculate surface area and volume of 3D shapes

CODE:

```
interface Shape3D {
double volume();
double surfaceArea(); }
class Sphere implements Shape3D {
 double radius;

Sphere(double radius) {
   this.radius = radius;
}

public double volume() {
   return (4.0 / 3.0) * Math.PI * Math.pow(radius, 3);
}

public double surfaceArea() {
   return 4 * Math.PI * Math.pow(radius, 2);
}
}
class Cube implements Shape3D {
double side;

Cube(double side) {
   this.side = side;
}
public double volume() {
   return Math.pow(side, 3);
}
public double surfaceArea() {
   return 6 * Math.pow(side, 2);
}


}
public class Record{
public static void main(String[] args) {
 Shape3D sphere = new Sphere(7.0); Shape3D cube = new Cube(6.0);
```

```
   System.out.println("Sphere Volume: " + sphere.volume());
    System.out.println("Sphere Surface Area: " + sphere.surfaceArea());
    System.out.println("Cube Volume: " + cube.volume());
    System.out.println("Cube Surface Area: " + cube.surfaceArea());
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Sphere Volume: 1436.7550402417319
Sphere Surface Area: 615.7521601035994
Cube Volume: 216.0
Cube Surface Area: 216.0
```

12D: To calculate balance after withdraw and deposit

CODE:

```
interface Bank {
    void details();
    void withdraw(double amount);
    void deposit(double amount);
}

class Account implements Bank {
    private String Name;
    private double bal;
    private String AccNo;

    Account(String Name, double bal, String AccNo) {
        this.Name = Name;
        this.bal = bal;
        this.AccNo = AccNo;
    }


    public void details() {
        System.out.println("Name: " + Name);
        System.out.println("Account Number: " + AccNo);
        System.out.println("Balance: " + bal);
```

```
    }

    public void withdraw(double amount) {
        if (bal >= amount) {
            bal -= amount;
            System.out.println("Balance after withdrawal: " + bal);
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    public void deposit(double amount) {
        bal += amount;
        System.out.println("Balance after deposit: " + bal);
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Name: Sahithi
Account Number: 123
Balance: 20000.0
Balance after deposit: 22000.0
Balance after withdrawal: 17000.0
```

ABSTRACT CLASS PROGRAMS:

13A: Calculate perimeter and area of shapes

CODE:

```
import java.util.Scanner;
abstract class Shape {
abstract double area();
abstract double perimeter();
}
class Square extends Shape{
double side;
Square(double side){
this.side = side;
}
double area(){
return (side * side);
}
double perimeter(){
return (4 * side);
}
}
class Rectangle extends Shape {
double length;
double breadth;
Rectangle(double length, double breadth) {
this.length = length;
this.breadth = breadth;
}
double area(){
return (length * breadth);
}
double perimeter(){
return (2 * (length + breadth));
}
}
public class Record{
public static void main(String[] args) {
Scanner s = new Scanner(System.in);
System.out.println("Enter the side of the square: ");
```

```
double side = s.nextDouble();
System.out.println("Enter length of the rectangle: ");
double l = s.nextDouble();
System.out.println("Enter the breadth of the rectangle: ");
double b = s.nextDouble();
Square sq = new Square(side);
System.out.println("Square:");
System.out.println("Area: " + sq.area());
System.out.println("Perimeter: " + sq.perimeter());
System.out.println();
Rectangle rec = new Rectangle(l, b);
System.out.println("Rectangle:");
System.out.println("Area: " + rec.area());
System.out.println("Perimeter: " + rec.perimeter());
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the side of the square:
10
Enter length of the rectangle:
10
Enter the breadth of the rectangle:
20
Square:
Area: 100.0
Perimeter: 40.0

Rectangle:
Area: 200.0
Perimeter: 60.0
```

13B: Add bonus to the salary and display employee details

CODE:

```
import java.util.Scanner;
abstract class Employee {
double salary;
String name;
int empId;
Employee(double salary, String name, int empId) {
this.salary = salary;
this.name = name;
this.empId = empId;
}
abstract void Salary(double bonus);
abstract void displayDetails();
}
class Manager extends Employee {
int teamSize;
Manager(double salary, String name, int empId, int teamSize) {
super(salary, name, empId);
this.teamSize = teamSize;
}
void Salary(double bonus){
System.out.println("Base Salary: " + salary);
System.out.println("Bonus: " + bonus);
System.out.println("Total Salary: " + (salary + bonus));
}
void displayDetails() {
System.out.println("Name: " + name);
System.out.println("Employee ID: " + empId);
System.out.println("Team Size: " + teamSize);
}
}
class Programmer extends Employee {
String language;
Programmer(double salary, String name, int empId, String language) {
super(salary, name, empId);
this.language = language;
}
void Salary(double bonus) {
```

```
System.out.println("Base Salary: " + salary);
System.out.println("Bonus: " + bonus);
System.out.println("Total Salary: " + (salary + bonus));
}
void displayDetails() {
System.out.println("Name: " + name);
System.out.println("Employee ID: " + empId);
System.out.println("Language Known: " + language);
}
}

public class Record{
public static void main(String[] args) {
Scanner s = new Scanner(System.in);
System.out.println("Name of the Manager: ");
String n1 = s.nextLine();
System.out.println("Enter Employee ID: ");
int i1 = s.nextInt();
System.out.println("Enter Team Size: ");
int t = s.nextInt();
System.out.println("Enter Salary of the Manager: ");
double sal1 = s.nextDouble();
System.out.println("Enter Bonus for Manager: ");
double b1 = s.nextDouble();
s.nextLine();
System.out.println("Name of the Programmer: ");
String n2 = s.nextLine();
System.out.println("Enter Employee ID: ");
int i2 = s.nextInt();
s.nextLine();
System.out.println("Enter Language Known: ");
String l = s.nextLine();
System.out.println("Enter Salary of the Programmer: ");
double sal2 = s.nextDouble();
System.out.println("Enter Bonus for Programmer: ");
double b2 = s.nextDouble();
Manager m = new Manager(sal1, n1, i1, t);
Programmer p = new Programmer(sal2, n2, i2, l);
System.out.println("\nManager Details:");
m.displayDetails();
```

```
m.Salary(b1);
System.out.println("\nProgrammer Details:");
 p.displayDetails();
 p.Salary(b2);
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Name of the Manager:
Sahithi
Enter Employee ID:
123
Enter Team Size:
6
Enter Salary of the Manager:
20000
Enter Bonus for Manager:
5000
Name of the Programmer:
Nagasai
Enter Employee ID:
124
Enter Language Known:
Python
Enter Salary of the Programmer:
25000
Enter Bonus for Programmer:
3000

Manager Details:
Name: Sahithi
Employee ID: 123
Team Size: 6
Base Salary: 20000.0
Bonus: 5000.0
Total Salary: 25000.0

Programmer Details:
Name: Nagasai
Employee ID: 124
Language Known: Python
Base Salary: 25000.0
Bonus: 3000.0
Total Salary: 28000.0
```

13C: Calculate surface area and volume of 3D shapes

CODE:

```
abstract class Shape3D {
 public abstract double volume();
 public abstract double surfaceArea();
}
class Sphere extends Shape3D {
double radius;
Sphere(double radius) {
    this.radius = radius;
}
public double volume() {
    return (4.0 / 3.0) * Math.PI * Math.pow(radius, 3);
}
public double surfaceArea() {
    return 4 * Math.PI * Math.pow(radius, 2);
}
}
class Cube extends Shape3D {
 double side;
Cube(double side) {
    this.side = side;
}
public double volume() {
    return Math.pow(side, 3);
}
 public double surfaceArea() {
    return 6 * Math.pow(side, 2);
}
}
public class Record {
 public static void main(String[] args) {
 Shape3D sphere = new Sphere(7.0);
 Shape3D cube = new Cube(6.0);
   System.out.println("Sphere Volume: " + sphere.volume());
    System.out.println("Sphere Surface Area: " + sphere.surfaceArea());
    System.out.println("Cube Volume: " + cube.volume());
    System.out.println("Cube Surface Area: " + cube.surfaceArea());
}
```

```
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Sphere Volume: 1436.7550402417319
Sphere Surface Area: 615.7521601035994
Cube Volume: 216.0
Cube Surface Area: 216.0
```

13D: To calculate balance after withdraw and deposit

CODE:

```
abstract class Bank {
    public abstract void details();
    public abstract void withdraw(double amount);
    public abstract void deposit(double amount);
}

class Account extends Bank {
    private String Name;
    private double bal;
    private String AccNo;

    Account(String Name, double bal, String AccNo) {
        this.Name = Name;
        this.bal = bal;
        this.AccNo = AccNo;
    }

    public void details() {
        System.out.println("Name: " + Name);
        System.out.println("Account Number: " + AccNo);
        System.out.println("Balance: " + bal);
    }

    public void withdraw(double amount) {
        if (bal >= amount) {
            bal -= amount;
            System.out.println("Balance after withdrawal: " + bal);
```

```
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    public void deposit(double amount) {
        bal += amount;
        System.out.println("Balance after deposit: " + bal);
    }
}

public class Record {
    public static void main(String[] args) {
        Account a = new Account("Sahithi", 20000, "123");
        a.details();
        a.deposit(2000);
        a.withdraw(5000);
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Name: Sahithi
Account Number: 123
Balance: 20000.0
Balance after deposit: 22000.0
Balance after withdrawal: 17000.0
```

ENCAPSULATION

14A: To calculate area and perimeter of rectangle.

CODE:
```
import java.util.Scanner;
class Rectangle {
   private double length;
   private double width;

public double getlength(){
return length;
}
public double getwidth(){
return width;
}
public void setlength(double length){
this.length = length;
}
public void setwidth(double width){
this.width = width;
}
public void Area() {
System.out.println("Area: " + (length * width));
}
public void perimeter(){
System.out.println("Perimeter: "+ 2 * (length + width));
}
}

public class Record {
public static void main(String[] args) {
Scanner s = new Scanner(System.in);
Rectangle r = new Rectangle();
System.out.println("Enter the length: ");
double length= s.nextDouble();
System.out.println("Enter the width: ");
double width= s.nextDouble();
r.setlength(length);
```

```
r.setwidth(width);
double t=r.getlength();
double u=r.getwidth();
r.Area();
r.perimeter();
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the length:
10
Enter the width:
10
Area: 100.0
Perimeter: 40.0
```

14B: To calculate total amount per unit area of house.

CODE:

```
import java.util.Scanner;

class House {
    private double Area;
public double getArea(){
return Area;
}
public void setArea(double Area){
this.Area = Area;
}
public void displayArea(double Area){
System.out.println("Area of the house: "+ Area);
}
public void Cost(double Area) {
System.out.println("Total Amount: "+Area*100);
}
}
public class Record {
public static void main(String[] args) {
```

```
Scanner s = new Scanner(System.in);
House h = new House();
System.out.println("Enter the Area: ");
double Area= s.nextDouble();
h.setArea(Area);
double a=h.getArea();
h.displayArea(Area);
h.Cost(Area);
    }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the Area:
1200
Area of the house: 1200.0
Total Amount: 120000.0
```

14C: To calculate balance after withdrawal and deposit

CODE:

```
import java.util.Scanner;
class Bank{
private String Accno;
private String Name;
private double bal;
public String getAccno(){
return Accno;
}
public String getName(){
return Name;
}
public double getbal(){
return bal;
}
public void setAccno(String Accno){
this.Accno=Accno;
```

```
}
public void setName(String Name){
this.Name=Name;
}
public void setbal(double bal){
this.bal=bal;
}
public void withdraw(double amount){
if (bal>0){
bal-=amount;
System.out.println("Amount after withdrawal: "+ bal);
}
else{
System.out.println("Insufficient Balance");
}
}
public void deposit(double amount){
bal+=amount;
System.out.println("Amount after deposit: "+bal);
}
public void displayDetails(){
System.out.println("Account no: "+ Accno);
System.out.println("Name: "+Name);
System.out.println("Balance: "+bal);
}
}
public class Record{
public static void main(String[] args){
Scanner s= new Scanner(System.in);
Bank b= new Bank();
System.out.println("Enter your Account No.");
String Accno=s.nextLine();
System.out.println("Enter your Name: ");
String Name=s.nextLine();
System.out.println("Enter the balance");
double bal=s.nextDouble();
System.out.println("Enter the amount: ");
double amount=s.nextDouble();
b.setAccno(Accno);
b.setName(Name);
```

```
b.setbal(bal);
String a=b.getAccno();
String n=b.getName();
double d=b.getbal();
b.withdraw(amount);
b.deposit(amount);
b.displayDetails();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter your Account No.
123
Enter your Name:
Sahithi
Enter the balance
2000
Enter the amount:
100
Amount after withdrawal: 1900.0
Amount after deposit: 2000.0
Account no: 123
Name: Sahithi
Balance: 2000.0
```

14D: To display book details and price

CODE:

```
import java.util.Scanner;

class Book{
private String book;
private String author;
private double price;
public String getbook(){
return book;
```

```
}
public String getauthor(){
return author;
}
public double getprice(){
return price;
}
public void setbook(String book){
this.book=book;
}
public void setauthor(String author){
this.author=author;
}
public void setprice(double price){
this.price=price;
}
public void applydiscount(double percentage){
price-=(price*(percentage/100));
System.out.println("Price after discount: "+price);
}
}


Public class Record{
public static void main (String[] args){
Scanner s= new Scanner(System.in);
Book b= new Book();
System.out.println("Enter book name: ");
String book=s.nextLine();
System.out.println("Enter author name: ");
String author=s.nextLine();
System.out.println("Enter the price:");
double price=s.nextDouble();
System.out.println("Enter the discount percentage: ");
double percentage=s.nextDouble();
b.setbook(book);
b.setauthor(author);
b.setprice(price);
String B=b.getbook();
String A=b.getauthor();
```

```
double P=b.getprice();
b.applydiscount(percentage);
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter book name:
GGGM
Enter author name:
Holly Jackson
Enter the price:
400
Enter the discount percentage:
20
Price after discount: 320.0
```

PACKAGES

15A: To calculate area of rectangle (User-defined package)

CODE:

```
package Area;
public class Rectangle {
int l;
int b;
public Rectangle(int l, int b) {
this.l = l;
this.b = b;
}
public void area(){
System.out.println("Area is: " + (l * b));
}
public void perimeter() {
System.out.println("Perimeter is: " + (2 * (l + b)));
}
}

import Area.Rectangle;
public class Record {
public static void main(String[] args) {
Rectangle r = new Rectangle(5, 10);
r.area();
r.perimeter();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac -d . Rectangle.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Area is: 50
Perimeter is: 30
```

15B: To display account details (User-defined package)

CODE:

```
package Bank;
public class Account{
public String name;
public String AccNo;
public double bal;
public Account(String name, String AccNo, double bal) {
this.name = name;
this.AccNo = AccNo;
this.bal = bal;
}
public void displayDetails(){
System.out.println("Name: " + name);
System.out.println("Account Number: " + AccNo);
System.out.println("Balance: " + bal);
}
public void withdraw(double amount) {
if (bal >= amount) {
bal -= amount;
System.out.println("Balance after withdrawal: " + bal);
} else {
System.out.println("Insufficient Balance");
}
}
public void deposit(double amount) {
bal += amount;
System.out.println("Balance after deposit: " + bal);
}
}
import Bank.Account;
public class Record {
public static void main(String[] args) {
Account a = new Account("Sahithi", "123", 30000);
a.displayDetails();
a.withdraw(3000);
a.deposit(10000);
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac -d . Account.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Name: Sahithi
Account Number: 123
Balance: 30000.0
Balance after withdrawal: 27000.0
Balance after deposit: 37000.0
```

15C: To calculate balance after withdrawal and deposit (built-in packages)

CODE:

```
package Bank;
public class Account{
public String name;
public String AccNo;
public double bal;
public Account(String name, String AccNo, double bal) {
this.name = name;
this.AccNo = AccNo;
this.bal = bal;
}
public void displayDetails(){
System.out.println("Name: " + name);
System.out.println("Account Number: " + AccNo);
System.out.println("Balance: " + bal);
}
public void withdraw(double amount) {
if (bal >= amount) {
bal -= amount;
System.out.println("Balance after withdrawal: " + bal);
} else {
System.out.println("Insufficient Balance");
}
}
public void deposit(double amount) {
bal += amount;
System.out.println("Balance after deposit: " + bal);
```

```
}
}
import Bank.Account;
import java.util.*;
import java.io.*;

public class Record {
public static void main(String[] args) throws IOException {
Scanner sc = new Scanner(System.in);
BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
System.out.print("Enter Name: ");
String name = br.readLine();
System.out.print("Enter Account Number: ");
String accNo = sc.next();
System.out.print("Enter Initial Balance: ");
double balance = sc.nextDouble();
Account acc = new Account(name, accNo, balance);
acc.displayDetails();
System.out.print("Enter amount to withdraw: ");
double withdrawAmount = sc.nextDouble();
acc.withdraw(withdrawAmount);
System.out.print("Enter amount to deposit: ");
double depositAmount = sc.nextDouble();
acc.deposit(depositAmount);
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter Name: Sahithi
Enter Account Number: 123
Enter Initial Balance: 30000
Name: Sahithi
Account Number: 123
Balance: 30000.0
Enter amount to withdraw: 10000
Balance after withdrawal: 20000.0
Enter amount to deposit: 4000
Balance after deposit: 24000.0
```

15D: To calculate perimeter and area of Rectangle (built-in packages)

CODE:

```
package Area;
public class Rectangle {
int l;
int b;
public Rectangle(int l, int b) {
this.l = l;
this.b = b;
}
public void area(){
System.out.println("Area is: " + (l * b));
}
public void perimeter() {
System.out.println("Perimeter is: " + (2 * (l + b)));
}
}
import Area.Rectangle;
import java.util.*;
import java.io.*;

public class Record {
public static void main(String[] args) throws IOException {
Scanner sc = new Scanner(System.in);
System.out.print("Enter length of the rectangle: ");
int length = sc.nextInt();
System.out.print("Enter breadth of the rectangle: ");
int breadth = sc.nextInt();
Rectangle r = new Rectangle(length, breadth);
r.area();
r.perimeter();
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter length of the rectangle: 20
Enter breadth of the rectangle: 10
```

EXCEPTION HANDLING

16A: Arithmetic Exception

CODE:

```
import java.util.Scanner;
class Record {
public static void main(String[] args) {
Scanner scanner = new Scanner(System.in);
System.out.print("Enter the numerator: ");
int n = scanner.nextInt();
System.out.print("Enter the denominator: ");
int m = scanner.nextInt();
try {
int ans = n / m;
System.out.println("Answer: " + ans);
}
catch (ArithmeticException e) {
System.out.println("Error: Division by zero is not allowed!");
}
finally {
scanner.close();
System.out.println("Program continues after handling the exception.");
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter the numerator: 10
Enter the denominator: 0
Error: Division by zero is not allowed!
Program continues after handling the exception.
```

16B: Illegal Argument Exception

CODE:

```
import java.util.Scanner;
public class Record{
public static void main(String[] args) {
Scanner scanner = new Scanner(System.in);
System.out.print("Enter a number: ");
int n = scanner.nextInt();
trynumber(n);
}
public static void trynumber(int n) {
try {
checkEvenNumber(n);
System.out.println(n + " is even.");
} catch (IllegalArgumentException e) {
System.out.println("Error: " + e.getMessage());
}
}
public static void checkEvenNumber(int number) {
if (number % 2 != 0) {
throw new IllegalArgumentException(number + " is odd.");
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Enter a number: 13
Error: 13 is odd.
```

16C: String Index Out of Bound Exception

CODE:

```
class Record{
public static void main(String args[]) {
try {
String a = "Java File Handling ";
char c = a.charAt(24);
System.out.println(c);
}
catch (StringIndexOutOfBoundsException e) {
System.out.println("StringIndexOutOfBoundsException occurred!");
}
finally {
System.out.println("Execution completed");
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
StringIndexOutOfBoundsException occurred!
Execution completed
```

16D:Null Pointer Exception

CODE:

```
public class Record{

public static void main(String[] args) {

try {
String str = null;
System.out.println(str.length());
} catch (NullPointerException e) {
```

```
System.out.println("Error: Cannot access length of a null string.");
} finally {
System.out.println("Execution completed.");
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Error: Cannot access length of a null string.
Execution completed.
```

FILE HANDLING

17A:Creating a file

CODE:

```
import java.io.File;
import java.io.IOException;
public class Record{
public static void main(String[] args) {
try {
File myObj = new File("record.txt");
if (myObj.createNewFile()) {
System.out.println("File created: " + myObj.getName());
} else {
System.out.println("File already exists.");
}
} catch (IOException e) {
System.out.println("An error occurred.");
 e.printStackTrace();
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
File created: record.txt
```

17B: Write a file

CODE:

```
import java.io.FileWriter;
import java.io.IOException;
public class Record {
public static void main(String[] args) {
try {
```

```
FileWriter myWriter = new FileWriter("record.txt");
myWriter.write("Java File Handling");
myWriter.close();
System.out.println("Successfully wrote to the file.");
} catch (IOException e) {
System.out.println("An error occurred.");
e.printStackTrace();
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Successfully wrote to the file.
```

17C: Read a file

CODE:

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;
public class Record{
public static void main(String[] args) {
try {
File myObj = new File("record.txt");
Scanner myReader = new Scanner(myObj);
while (myReader.hasNextLine()) {
String data = myReader.nextLine();
System.out.println(data);
}
myReader.close();
} catch (FileNotFoundException e) {
System.out.println("An error occurred.");
e.printStackTrace();
}
}
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Java File Handling
```

17D: Delete a file

CODE:

```
import java.io.File;
public class Record{
public static void main(String[] args) {
File myObj = new File("record.txt");
if (myObj.delete()) {
System.out.println("Deleted the file: " + myObj.getName());
 } else {
 System.out.println("Failed to delete the file.");
}
 }
}
```

OUTPUT:

```
C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>javac Record.java

C:\Users\LOMADA SAHITHI\OneDrive\Desktop\java>java Record
Deleted the file: record.txt
```